Exercise list R2b — Bond valuation, the term structure and the cost of debt

Worked solutions

Statements are in `Listas/lista-r2b-bond-valuation.md`; each exercise is cited by number and printed page and is not restated. Short answers: `lista-r2b-bond-valuation-solutions-short.pdf`. Editions are the local copies: Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013); Higgins = Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed.; Grinblatt–Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002). Ross writes decimals with commas and \$1.000 for one thousand; these solutions use points. Numbers labelled "illustration" are not part of the statement.

Formulas used throughout. With coupon C per period, face value F, n periods and yield y per period,

$$P = C \cdot A(y,n) + \frac{F}{(1+y)^n}, \qquad A(y,n) = \frac{1-(1+y)^{-n}}{y} \quad \text{(annuity factor)}, \tag{1}$$

$$P = \sum_{t=1}^{n} \frac{CF_t}{(1+r_t)^t} \quad \text{(spot rates } r_t\text{, Berk Eq. 6.4)}. \tag{2}$$

A semiannual bond quoted at an APR y^{APR} has $y = y^{\text{APR}}/2$ per period and $C =$ coupon rate$\times F/2$.

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1. Zero-coupon prices and the yield curve — Berk 6.4, p. 172

What is asked. (a) The yield to maturity of each of the five default-free zeros from its price per \$100 of face value. (b) A plot of yield against maturity. (c) Whether the curve slopes up, down or is flat.

Approach. A zero-coupon bond has a single cash flow, the face value at maturity, so its price is that face value discounted at its own yield: $P_n = 100/(1+YTM_n)^n$. Inverting gives the yield. The zero-coupon yields are the *spot rates* of block 2A.1, the discount rates used to price any other default-free bond.

Notation.

- $n \in \{1,\dots,5\}$: maturity in years; P_n: price per \$100 face value; YTM_n: annual yield (spot rate) for maturity n.

Step 1 — Invert the pricing formula. The price of a zero is

$$P_n = \frac{100}{(1+YTM_n)^n} \implies (1+YTM_n)^n = \frac{100}{P_n} \implies YTM_n = \left(\frac{100}{P_n}\right)^{1/n} - 1. \tag{3}$$

Step 2 — Substitute each price. We compute the ratio $100/P_n$, take the nth root and subtract one.

n	P_n	$100/P_n$	$(100/P_n)^{1/n}$	YTM_n
1	95.51	1.047011	$1.047011^{1} = 1.047011$	4.70%
2	91.05	1.098298	$1.098298^{1/2} = 1.047997$	4.80%
3	86.38	1.157675	$1.157675^{1/3} = 1.050015$	5.00%
4	81.65	1.224740	$1.224740^{1/4} = 1.051988$	5.20%
5	76.51	1.307019	$1.307019^{1/5} = 1.055009$	5.50%

Answer (a). $YTM_1 = 4.70\%$, $YTM_2 = 4.80\%$, $YTM_3 = 5.00\%$, $YTM_4 = 5.20\%$, $YTM_5 = 5.50\%$.

Step 3 — Plot. The curve joins the points $(1, 4.70)$, $(2, 4.80)$, $(3, 5.00)$, $(4, 5.20)$, $(5, 5.50)$ (maturity in years, yield in %).

Answer (b). A rising line through those five points, steepening after year 3.

Step 4 — Slope. Each yield exceeds the previous one: $4.70 < 4.80 < 5.00 < 5.20 < 5.50$.

Answer (c). Upward sloping.

Intuition. Longer zeros are cheaper per year of waiting than a constant rate would imply: the market asks a higher annual rate to lend for longer. These five numbers are all one needs to price any default-free bond paying on those dates (Exercise 8).

Common mistakes.

- Computing $(100 - P_n)/P_n$ and calling it the annual yield: that is the *total* return over n years, not an annual rate.
- Dividing the total return by n (simple interest) instead of taking the nth root.

Ref: Berk §6.2 "Zero-Coupon Bonds", pp. 147–151.

2. Holding-period return on a bond — Higgins 5.5, p. 188

What is asked. (a) The return over one year on a bond bought at \$1,050 and worth \$940 a year later, after one annual coupon at 6% of \$1,000. (b) Two reasons why its price could have fallen.

Approach. The holding-period return is the definition given in the statement: income plus capital gain over the starting price. Part (b) uses the pricing formula (1): the price falls if the yield the market demands rises, either because riskless rates rise or because the issuer's credit risk rises.
Notation.

- $P_0 = 1{,}050$, $P_1 = 940$: prices at the start and end of the year; I: coupon received; HPR: holding-period return.

Step 1 — Coupon. The annual coupon is

$$I = 0.06 \times 1{,}000 = 60. \tag{4}$$

Step 2 — Return. We add the coupon to the price change and divide by the starting price:

$$HPR = \frac{I + (P_1 - P_0)}{P_0} = \frac{60 + (940 - 1{,}050)}{1{,}050} = \frac{60 - 110}{1{,}050} = \frac{-50}{1{,}050} = -4.76\%. \tag{5}$$

Answer (a). $HPR = -4.76\%$.

Step 3 — Why the price fell. From (1), with coupon and face fixed, P falls only if the discount rate y rises (the passage of time alone pulls a premium bond toward par, but a 6% bond at \$1,050 would only lose a few dollars that way, not \$110).

Answer (b). (i) Market interest rates rose (interest-rate risk); (ii) the market judged the issuer riskier (a downgrade or worse prospects raised the default premium). Either raises the yield and lowers the price.

Intuition. The coupon is fixed, so all news reaches the bondholder through the price. Even a default-free bond has a risky one-year return unless it matures exactly at the end of the holding period.

Common mistakes.

- Forgetting the coupon: $(940 - 1{,}050)/1{,}050 = -10.5\%$ is the capital loss only.
- Dividing by the face value \$1,000 instead of the price paid.

Book answer. Matches the book's answer, −4.76 percent, with interest rates or perceived risk rising as the reasons (p. 410).
Ref: Higgins Ch. 5, "Bonds" and "Bond Ratings", pp. 143–150; Ch. 7 for bond pricing.

3. Price of a seasoned semiannual bond — Ross 7.6, p. 232

What is asked. The price today of a bond issued one year ago with an 11-year maturity, a 6.9% coupon paid twice a year, when its yield to maturity is 7.4% (APR, semiannual).

Approach. Ross's bond value formula (1) with half-year periods. Two conversions do the work: the remaining life is $11 - 1 = 10$ years, i.e. 20 half-years, and both the coupon and the quoted yield are halved. Face value \$1,000 (the book's convention; see the Context in the list).
Notation.

- $F = 1{,}000$; $C = 34.5$: coupon per half-year; $y = 3.7\%$: yield per half-year; $n = 20$ half-years.

Step 1 — Per-period inputs. We convert the annual figures to half-years:

$$C = \frac{0.069 \times 1{,}000}{2} = \frac{69}{2} = 34.5, \qquad y = \frac{0.074}{2} = 0.037, \qquad n = 2 \times (11 - 1) = 20. \tag{6}$$

Step 2 — Factors. The compounding factor and the annuity factor are

$$(1.037)^{20} = 2.068117, \qquad \frac{1}{(1.037)^{20}} = 0.483532, \tag{7}$$

$$A(0.037, 20) = \frac{1 - 0.483532}{0.037} = \frac{0.516468}{0.037} = 13.958605. \tag{8}$$

Step 3 — Price. Substituting into (1):

$$P = (34.5)(13.958605) + (1{,}000)(0.483532) = 481.57 + 483.53 = 965.10. \tag{9}$$

Answer. $P = \$965.10$ per \$1,000 of face value (a discount bond: coupon 6.9% < yield 7.4%).

Intuition. The bond pays 6.9% when the market wants 7.4%, so it must sell below par; the discount (\$34.90) is the present value of the coupon shortfall $(37-34.5)\times A(0.037, 20) = 2.5\times 13.958605 = 34.90$.

Common mistakes.

- Using 11 periods or 22 half-years: the bond is one year old, so 10 years (20 half-years) remain.
- Discounting semiannual coupons at 7.4% per period, or annual coupons at 3.7%.

Ref: Ross §7.1 "Valores e retornos dos títulos de dívida", bond value formula and Example 7.1, pp. 192–196.

4. Callable vs. noncallable bonds — Higgins 5.4, p. 188

What is asked. Whether two otherwise identical 20-year bonds, one callable at \$1,000 after five years and one not, have different yields; which is higher; and why.

Approach. A call provision is an option held by the *issuer*: the bondholder has sold it. By the law of one price, the callable bond is a straight bond minus a call option, so its price is lower and, for the same promised cash flows, its yield is higher.

Notation.

- P^{straight}, P^{callable}: prices; $V^{\text{call}} \geq 0$: value of the issuer's option to repay at \$1,000.

Step 1 — Decompose the callable bond. The holder of a callable bond owns the straight bond and is short the call:

$$P^{\text{callable}} = P^{\text{straight}} - V^{\text{call}} \leq P^{\text{straight}}. \tag{10}$$

Step 2 — When is the call exercised. The issuer calls when rates have fallen so that the straight bond is worth more than \$1,000: it refinances at the lower rate. The bondholder loses exactly the gain she would otherwise have had and must reinvest at the lower rate. The call caps her upside.

Step 3 — From price to yield. Both bonds promise the same coupons and face value if not called; a lower price for the same promised flows means a higher discount rate solving (1).

Answer. Yes: the callable bond's yield to maturity is *higher*, compensating investors for the option they give the issuer (the call is exercised precisely when it hurts them: after a fall in rates).

Intuition. Exercise 14 puts a number on V^{call}: about 12% of face value for a Treasury callable within a year. The yield difference is a price for an option, not a default premium.

Common mistakes.

- Thinking the callable bond is *safer* because it may be repaid early; early repayment happens when the investor least wants it.
- Confusing the call (issuer's option) with a put (holder's option, which lowers the yield).

Ref: Higgins Ch. 5, "Bonds" (call provisions), p. 145; Ross §7.2 "Opção de resgate antecipado", p. 207.

5. Expected return vs. yield on a corporate bond — Berk Critical Thinking 8, p. 172

What is asked. Why the expected return of a corporate bond differs from its yield to maturity.

Approach. Block 2A.10: the yield to maturity is the internal rate of return of the *promised* cash flows, the expected return is the rate that equates the price to the *expected* cash flows. With default risk the expected cash flows are smaller, so the same price implies a smaller rate.

Notation.

- P: price; F: promised payment (one-year zero, for clarity); p: default probability; C: recovery in default.
- y: promised yield, $P = F/(1+y)$; r_D: expected return (cost of debt), $P = [F(1-p)+Cp]/(1+r_D)$.

Step 1 — Two definitions, one price. From the two pricing equations,

$$1 + y = \frac{F}{P}, \qquad 1 + r_D = \frac{F(1-p)+Cp}{P}. \tag{11}$$

Step 2 — Compare. Since $C < F$, $F(1-p)+Cp = F - p(F-C) < F$ whenever $p > 0$, hence

$$r_D < y. \tag{12}$$

Step 3 — Illustration. $F = 100$, $P = 90$, $p = 5\%$, $C = 40$:

$$y = \frac{100}{90} - 1 = 11.11\%,$$

$$\mathbb{E}[\text{payoff}] = (0.95)(100) + (0.05)(40) = 95 + 2 = 97, \qquad r_D = \frac{97}{90} - 1 = 7.78\%.$$

Answer. The yield is computed from the promised payments, which are the most the holder can receive; with a positive probability of default and partial recovery the expected payments are lower, so the expected return is below the yield. The yield spread over Treasuries pays for expected default losses *and* a risk premium; only the latter adds to the expected return.

Intuition. Using the yield as r_D in a WACC overstates the cost of debt, and the error is largest for junk debt, exactly when default is most likely (Berk Ch. 13 cites KB Homes: 8.6% promised, about 7.3% expected).

Common mistakes.

- Treating a high yield as a high expected return: a bond close to default can promise 20% and be expected to earn far less.
- Using the coupon rate as the cost of debt; it is historical and is neither the yield nor the expected return.

Ref: Berk §6.5 “Corporate Bond Yields”, p. 166; Ch. 13 box “Using the Coupon Rate as the Cost of Debt”, p. 387; Ricca 2A.10 (slides 103–105).

6. Credit spreads on one-year zeros — Berk 6.24, p. 174

What is asked. (a) The price, in % of face, of a one-year AAA zero yielding 3.2%. (b), (c) The credit spreads of AAA and B bonds over the Treasury. (d) How spreads move with the rating, and why.

Approach. One-year zeros: price $= 100/(1+y)$. The credit (default) spread is the difference between a corporate yield and the Treasury yield of the same maturity.

Notation.

- $y^T = 3.1\%$, $y^{AAA} = 3.2\%$, $y^{BBB} = 4.2\%$, $y^B = 4.9\%$; spread $s^k = y^k - y^T$.

Step 1 — Price.

$$P^{AAA} = \frac{100}{1 + 0.032} = \frac{100}{1.032} = 96.90. \tag{13}$$

Answer (a). 96.90% of face value.

Step 2 — Spreads.

$$s^{AAA} = 3.2 - 3.1 = 0.1\%, \qquad s^{BBB} = 4.2 - 3.1 = 1.1\%, \qquad s^B = 4.9 - 3.1 = 1.8\%. \tag{14}$$

Answer (b). 0.1% (10 basis points).

Answer (c). 1.8% (180 basis points).

Step 3 — Pattern. $0.1 < 1.1 < 1.8$: spreads rise as the rating falls.

Answer (d). The spread widens as the rating worsens, because lower-rated issuers have higher default probabilities (and lower expected recoveries): their promised yields must cover larger expected losses and a larger risk premium.

Intuition. The spread is the market price of credit risk. By Exercise 5, a B bond's 4.9% is a *promise*; its expected return is lower.

Common mistakes.

- Quoting spreads relative to AAA rather than to the Treasury.
- Reading the B spread as the expected extra return; part of it only compensates expected default losses.

Ref: Berk §6.5 "Corporate Bonds", "Corporate Bond Yields" and "Bond Ratings", pp. 164–167.

7. A bond's price over time at a constant yield — Berk 6.16–6.18, p. 173

What is asked. (16) The issue price of a 10-year, 7% annual-coupon \$1,000 bond at a 6% yield. (17) Its price one year later just before the first coupon, and (18) just after it, if the yield stays at 6%.

Approach. Formula (1). At a constant yield, the value of any bond grows at the yield between coupon dates (it is a present value rolling forward), and drops by the coupon when it is paid. The price "after" is just the price of a 9-year bond.

Notation.

- $C = 70$, $F = 1{,}000$, $y = 6\%$; P_0: issue price; P_1^-, P_1^+: prices just before and just after the first coupon.

Step 1 — (16) Issue price. The factors at 6% for 10 years are

$$A(0.06, 10) = \frac{1 - (1.06)^{-10}}{0.06} = \frac{1 - 0.558395}{0.06} = 7.360087, \qquad \frac{1}{(1.06)^{10}} = 0.558395, \tag{15}$$

so

$$P_0 = (70)(7.360087) + (1{,}000)(0.558395) = 515.21 + 558.39 = 1{,}073.60. \tag{16}$$

Answer (16). \$1,073.60 (a premium bond: coupon 7% > yield 6%).

Step 2 — (17) Just before the coupon. One year on, the same cash flows are one year closer; their present value is P_0 carried forward at 6%:

$$P_1^- = P_0(1+y) = (1{,}073.60)(1.06) = 1{,}138.02. \tag{17}$$

Answer (17). \$1,138.02.

Step 3 — (18) Just after the coupon. The holder of the bond now owns a 9-year bond. At 6% for 9 years:

$$A(0.06, 9) = \frac{1 - 0.591898}{0.06} = 6.801692, \qquad \frac{1}{(1.06)^9} = 0.591898,$$
$$P_1^+ = (70)(6.801692) + (1{,}000)(0.591898) = 476.12 + 591.90 = 1{,}068.02. \tag{18}$$

Check: $P_1^- = 70 + P_1^+ = 70 + 1{,}068.02 = 1{,}138.02$, as in (17).

Answer (18). \$1,068.02.

Intuition. Over the year the holder earns $0.06 \times 1{,}073.60 = 64.42$: a coupon of 70 less a capital loss of $1{,}073.60 - 1{,}068.02 = 5.58$. A premium bond loses value as it ages (it is "pulled to par"), but its return stays at the yield. The saw-tooth path (rise, drop by the coupon) is how prices move with time alone.

Common mistakes.

- Answering (17) and (18) with the same number: the coupon is paid between the two instants.
- Thinking the price stays at \$1,073.60 because the yield did not change; with constant yield the *return*, not the price, is constant.

Ref: Berk §6.4 "Time and Bond Prices", pp. 158–160.

8. Pricing a coupon bond with spot rates — Berk 6.10, p. 173

What is asked. The price of a \$1,000, 4% semiannual-coupon Treasury maturing in 1.5 years, given spot rates (semiannual APRs) of 1% (0.5 years), 1.1% (1 year) and 1.3% (1.5 years).

Approach. Block 2A.1: by the law of one price, a coupon bond is a portfolio of zeros, one per cash flow, so each cash flow is discounted at the spot rate of its own date (Berk Eq. 6.4, equation (2)). Using a single rate would mix three different prices of time.

Notation.

- Cash flows: $CF_1 = CF_2 = 0.04 \times 1{,}000/2 = 20$, $CF_3 = 20 + 1{,}000 = 1{,}020$ at half-years 1, 2, 3.
- Spot rates per half-year: $r_1 = 1\%/2 = 0.5\%$, $r_2 = 1.1\%/2 = 0.55\%$, $r_3 = 1.3\%/2 = 0.65\%$.

Step 1 — Compounding factors.

$$(1.0055)^2 = 1.011030, \qquad (1.0065)^3 = 1.019627. \tag{19}$$

Step 2 — Discount each cash flow at its own spot rate.

$$P = \frac{20}{1.005} + \frac{20}{(1.0055)^2} + \frac{1{,}020}{(1.0065)^3} = \frac{20}{1.005} + \frac{20}{1.011030} + \frac{1{,}020}{1.019627}$$
$$= 19.90 + 19.78 + 1{,}000.37 = 1{,}040.05. \tag{20}$$

Answer. $P =$ \$1,040.05.

Step 3 — The bond's own yield (for intuition). Solving $1{,}040.05 = 20/(1+y) + 20/(1+y)^2 + 1{,}020/(1+y)^3$ gives $2y = 1.295\%$ APR: between the 1-year and 1.5-year spot rates and close to the latter, because most of the value comes from the last cash flow.

Intuition. The yield to maturity of a coupon bond is a (value-weighted) average of the spot rates of its dates. With an upward-sloping curve it lies just below the spot rate of its maturity.

Common mistakes.

- Discounting everything at 1.3%: gives a slightly lower price because the early coupons are over-discounted.
- Using the annual APR as a per-period rate, or compounding the 1.5-year rate for 1.5 rather than 3 half-years.

Ref: Berk Ch. 6 Appendix B "Valuing a Coupon Bond Using Zero-Coupon Yields", pp. 178–181; Ricca 2A.1 (slides 2–7).

9. Coupon rate for a new issue sold at par — Ross 7.19, p. 233

What is asked. The coupon rate that makes new 20-year semiannual bonds sell at par, when the firm's existing 20-year, 8% semiannual bonds sell for \$930.

Approach. A bond sells at par exactly when its coupon rate equals its yield to maturity. The market's required return on 20-year debt of this issuer is the yield of the existing bonds, so: solve for that yield, then set the new coupon equal to it. This is the cost of new debt; the old 8% coupon is history.

Notation.

- Existing bond: $C = 40$ per half-year, $F = 1{,}000$, $n = 40$, $P = 930$; y: yield per half-year.

Step 1 — Equation for the yield. From (1),

$$930 = 40 \cdot \frac{1 - (1+y)^{-40}}{y} + \frac{1{,}000}{(1+y)^{40}}. \tag{21}$$

The bond is at a discount, so $y > 4\%$ per half-year.

Step 2 — Bracket and solve. Trial values: at $y = 4.5\%$ (9% APR) the right side is 907.99; at $y = 4.25\%$ (8.5% APR) it is 952.31. The root lies between and, by iteration (or a financial calculator), is $y = 4.3736\%$. Check:

$$A(0.043736, 40) = 18.738512, \qquad \frac{1}{(1.043736)^{40}} = 0.180460,$$
$$(40)(18.738512) + (1{,}000)(0.180460) = 749.54 + 180.46 = 930.00. \tag{22}$$

Step 3 — Annual quote. The yield is quoted as an APR:

$$y^{\text{APR}} = 2 \times 4.3736\% = 8.75\%, \qquad \text{EAR} = (1.043736)^2 - 1 = 8.94\%. \tag{23}$$

Answer. Set the coupon rate at 8.75% per year (4.375% per half-year); then the new bonds sell at par.

Intuition. The cost of new debt is the yield the market currently demands, not the coupon promised in the past (block 2A.10: r_D is not the coupon). The old bonds trade at a discount precisely because 8% is below what the market now asks.

Common mistakes.

- Answering 8% (the old coupon).
- Reporting the EAR 8.94% as the coupon rate; coupon rates, like yields, are quoted as APRs.

Ref: Ross §7.1 "Valores e retornos dos títulos de dívida", pp. 192–199.

10. Ratings, prices and issue size — Berk 6.26, p. 175

What is asked. (a) The price of a 5-year, 6.5% annual-coupon \$1,000 bond if rated AA. (b) The total face value HMK must issue to raise \$10 million at that price. (c) The rating at which the bond sells at par. (d) The rating implied by a price of \$959.54, and whether that is junk.

Approach. The rating maps to a yield (the table), the yield to a price via (1). A bond sells at par when its yield equals its coupon; and a price can be inverted to a yield and hence to a rating.

Notation.

- $C = 65$, $F = 1{,}000$, $n = 5$; y^k: yield for rating k (6.20, 6.30, 6.50, 6.90, 7.50% for AAA to BB).

Step 1 — (a) Price at AA ($y = 6.3\%$).

$$A(0.063, 5) = \frac{1 - (1.063)^{-5}}{0.063} = \frac{1 - 0.736773}{0.063} = 4.178207,$$
$$P^{AA} = (65)(4.178207) + (1{,}000)(0.736773) = 271.58 + 736.77 = 1{,}008.36. \tag{24}$$

Answer (a). \$1,008.36.

Step 2 — (b) Number of bonds. Proceeds per bond are \$1,008.36, so

$$\frac{10{,}000{,}000}{1{,}008.36} = 9{,}917.13 \;\to\; 9{,}917 \text{ bonds}, \qquad 9{,}917 \times 1{,}000 = \$9{,}917{,}000 \text{ of face value.} \tag{25}$$

(They raise $9{,}917 \times 1{,}008.36 = \$9{,}999{,}870.56$, within rounding of \$10 million.)

Answer (b). \$9.917 million of principal (9,917 bonds).

Step 3 — (c) Par. At $y = 6.5\%$ the coupon equals the yield, so each coupon exactly pays the required return and $P = 1{,}000$.

Answer (c). A rating (yield 6.50%).

Step 4 — (d) Implied rating. The price is below par, so the yield exceeds 6.5%. Try BB, $y = 7.5\%$:

$$A(0.075, 5) = \frac{1 - 0.696559}{0.075} = 4.045885, \qquad \frac{1}{(1.075)^5} = 0.696559,$$
$$P = (65)(4.045885) + (1{,}000)(0.696559) = 262.98 + 696.56 = 959.54. \tag{26}$$

The price matches, so the yield is 7.50%.

Answer (d). BB. Yes, they are junk bonds: BB is below the four investment-grade classes (AAA, AA, A, BBB).

Intuition. Each notch down the rating scale raises the yield and lowers what the firm raises per bond. For HMK, the difference between AA and BB is $1{,}008.36 - 959.54 = 48.82$ per bond, almost 5% of the issue.

Common mistakes.

- In (b), dividing \$10 million by \$1,000 face (10,000 bonds) instead of by the price.
- In (d), calling BBB junk: BBB is the lowest investment grade.

Ref: Berk §6.5 "Bond Ratings", Table 6.6 and "Corporate Yield Curves", pp. 166–167.

11. Interest-rate sensitivity: maturity and coupon — Berk 6.21–6.22, p. 174

What is asked. (21) The percentage price change of each of four bonds (two zeros, two coupon bonds; 10 and 15 years) when the yield falls from 6% to 5%. (22) Which is most and which least sensitive, and why.

Approach. Price each bond at both yields with (1) ($F = 1{,}000$; percentage changes do not depend on F). Sensitivity grows with the time at which value is received: longer maturity and lower coupon both push value later. The summary measure is Macaulay duration, the value-weighted average time of the cash flows.

Notation.

- $P(y)$: price at yield y; $\Delta\% = P(5\%)/P(6\%) - 1$; $D = \sum_t t\,PV(CF_t)/P$: Macaulay duration at 6%.

Step 1 — Factors. $A(0.06, 15) = 9.712249$, $A(0.05, 15) = 10.379658$, $A(0.06, 10) = 7.360087$, $A(0.05, 10) = 7.721735$; $(1.06)^{-15} = 0.417265$, $(1.05)^{-15} = 0.481017$, $(1.06)^{-10} = 0.558395$, $(1.05)^{-10} = 0.613913$.

Step 2 — Prices and changes. For example, bond C:

$$\begin{aligned} P(6\%) &= (40)(9.712249) + (1{,}000)(0.417265) = 388.49 + 417.27 = 805.76, \\ P(5\%) &= (40)(10.379658) + (1{,}000)(0.481017) = 415.19 + 481.02 = 896.20, \\ \Delta\% &= 896.20/805.76 - 1 = 11.23\%. \end{aligned} \tag{27}$$

In full:

Bond	Coupon, mat.	$P(6\%)$	$P(5\%)$	$\Delta\%$	D (years)
A	0%, 15	$1{,}000 \times 0.417265 = 417.27$	$1{,}000 \times 0.481017 = 481.02$	+15.28%	15.00
B	0%, 10	$1{,}000 \times 0.558395 = 558.39$	$1{,}000 \times 0.613913 = 613.91$	+9.94%	10.00
C	4%, 15	$388.49 + 417.27 = 805.76$	$415.19 + 481.02 = 896.20$	+11.23%	11.11
D	8%, 10	$588.81 + 558.39 = 1{,}147.20$	$617.74 + 613.91 = 1{,}231.65$	+7.36%	7.45

Answer (21). A +15.28%, B +9.94%, C +11.23%, D +7.36%.

Step 3 — Ranking. $15.28 > 11.23 > 9.94 > 7.36$: A > C > B > D, which is the order of the durations $15 > 11.11 > 10 > 7.45$.

Answer (22). Most sensitive: A (15-year zero); least sensitive: D (10-year, 8% coupon). A receives all its value at year 15, where a change in the discount rate is compounded 15 times; D receives large coupons early, which are barely affected by the rate.

Intuition. The relevant "maturity" for interest-rate risk is duration, not the final date: C matures in 15 years like A but its coupons pull its duration to 11.1 years, so it is less sensitive than A and (because $11.1 > 10$) more sensitive than the 10-year zero B.

Common mistakes.

- Ranking by maturity alone (would put C level with A).
- Computing the change relative to the new price instead of the old one.

Ref: Berk §6.4 "Interest Rate Changes and Bond Prices", pp. 156–158 (duration, footnote 4, p. 161).

12. A bond with deferred, stepped payments — Ross 7.31 (DESAFIO), p. 234

What is asked. The price of bond M ($20,000 face, 20 years; nothing for 6 years, then $1,100 every six months for 8 years, then $1,400 every six months for 6 years, plus the face value at maturity) and of zero-coupon bond N ($20,000 face, 20 years), both at 7% compounded semiannually.

Approach. Value each block of level payments as an annuity at the date just before its first payment, then discount that value back to today (a deferred annuity); add the face value discounted 40 half-years. Bond N is that last term alone.

Notation.

- $r = 7\%/2 = 3.5\%$ per half-year; periods 1–12: no payment; 13–28: \$1,100 (16 payments); 29–40: \$1,400 (12 payments); $F = 20{,}000$ at 40.

Step 1 — Factors.

$$A(0.035, 16) = 12.094117, \quad A(0.035, 12) = 9.663334,$$
$$(1.035)^{12} = 1.511069, \quad (1.035)^{28} = 2.620172, \quad (1.035)^{40} = 3.959260. \tag{28}$$

Step 2 — First block (periods 13–28). Its value at period 12 is an ordinary 16-period annuity; we discount it 12 periods:

$$PV_1 = \frac{1{,}100 \times A(0.035, 16)}{(1.035)^{12}} = \frac{(1{,}100)(12.094117)}{1.511069} = \frac{13{,}303.53}{1.511069} = 8{,}804.05. \tag{29}$$

Step 3 — Second block (periods 29–40). Its value at period 28, discounted 28 periods:

$$PV_2 = \frac{1{,}400 \times A(0.035, 12)}{(1.035)^{28}} = \frac{(1{,}400)(9.663334)}{2.620172} = \frac{13{,}528.67}{2.620172} = 5{,}163.27. \tag{30}$$

Step 4 — Face value.

$$PV_3 = \frac{20{,}000}{(1.035)^{40}} = \frac{20{,}000}{3.959260} = 5{,}051.45. \tag{31}$$

Step 5 — Prices.

$$P_M = 8{,}804.05 + 5{,}163.27 + 5{,}051.45 = 19{,}018.78, \qquad P_N = 5{,}051.45. \tag{32}$$

Answer. $P_M = \$19{,}018.78$; $P_N = \$5{,}051.45$.

Intuition. Both bonds promise the same \$20,000 at year 20; M is worth \$13,967 more because of the coupons, even though none is paid for six years. Deferral is handled by a two-stage discount: annuity value at the start of the block, then a lump sum back to today.

Common mistakes.

- Discounting the first block's annuity value by 13 periods (its first payment date) instead of 12 (one period before it).
- Forgetting the \$20,000 face value of bond M.
- Using 7% per year with annual periods: the payments and the compounding are semiannual.

Ref: Ross §6.2 (annuities), pp. 155–163, and §7.1, pp. 192–199.

13. Spot and annuity yields from par yields — Grinblatt & Titman 10A.3, p. 369

What is asked. Given par yields of 4.5%, 5%, 5.25%, 5.25% for years 1–4 (annual coupons and compounding), compute the spot (zero-coupon) yields and the annuity yields for years 1–4.

Approach. Bootstrapping (G&T §10A.2): a par bond is worth 100 and pays its par yield as coupon. Starting from year 1 (a par bond of one year is a zero), strip the earlier coupons using the discount factors already found; what is left is a zero-coupon payment at the final date whose price gives the next discount factor. Annuity yields then follow from the discount factors: the annuity's price is the sum of the discount factors and its yield is the IRR.

Notation.

- c_t: par yield (coupon per 100) for maturity t; $d_t = 1/(1+s_t)^t$: discount factor; s_t: spot yield; $A_t = \sum_{k \le t} d_k$: price of a \$1 annuity; a_t: annuity yield, solving $A(a_t, t) = A_t$.

Step 1 — Bootstrap equation. For the t-year par bond, price 100 equals coupons plus final payment discounted with the spot factors:

$$100 = c_t \sum_{k=1}^{t-1} d_k + (100 + c_t)\, d_t \implies d_t = \frac{100 - c_t \sum_{k=1}^{t-1} d_k}{100 + c_t}, \qquad s_t = d_t^{-1/t} - 1. \tag{33}$$

Step 2 — Year 1. $d_1 = 100/104.5 = 0.956938$, so $s_1 = 4.50\%$.

Step 3 — Year 2. Strip the year-1 coupon: $5d_1 = 5 \times 0.956938 = 4.784689$.

$$d_2 = \frac{100 - 4.784689}{105} = \frac{95.215311}{105} = 0.906812, \qquad s_2 = (0.906812)^{-1/2} - 1 = 5.0126\%. \tag{34}$$

Step 4 — Year 3. Strip two coupons: $5.25(d_1 + d_2) = 5.25 \times 1.863750 = 9.784689$.

$$d_3 = \frac{100 - 9.784689}{105.25} = \frac{90.215311}{105.25} = 0.857153, \qquad s_3 = (0.857153)^{-1/3} - 1 = 5.2723\%. \tag{35}$$

Step 5 — Year 4. Strip three coupons: $5.25(d_1 + d_2 + d_3) = 5.25 \times 2.720903 = 14.284740$.

$$d_4 = \frac{100 - 14.284740}{105.25} = \frac{85.715260}{105.25} = 0.814397, \qquad s_4 = (0.814397)^{-1/4} - 1 = 5.2667\%. \tag{36}$$

Check: $5.25 \times 2.720903 + 105.25 \times 0.814397 = 14.284740 + 85.715260 = 100$.

Answer (spot). $s_1 = 4.50\%$, $s_2 = 5.01\%$, $s_3 = 5.27\%$, $s_4 = 5.27\%$ (5.2723% and 5.2667%: the 4-year spot is slightly *below* the 3-year).

Step 6 — Annuity prices. Summing the discount factors:

$$A_1 = 0.956938, \quad A_2 = 1.863750, \quad A_3 = 2.720903, \quad A_4 = 3.535300. \tag{37}$$

Step 7 — Annuity yields. Solve $\sum_{k=1}^{t} (1 + a_t)^{-k} = A_t$ for each t (one unknown, monotone, so iteration converges):

$$a_1 = 4.50\%, \quad a_2 = 4.8356\%, \quad a_3 = 5.0460\%, \quad a_4 = 5.1296\%. \tag{38}$$

For example $t = 2$: $1/1.048356 + 1/1.048356^2 = 0.953874 + 0.909876 = 1.863751 \approx A_2 = 1.863750$ (the gap is the rounding of a_2).

Answer (annuity). $a_1 = 4.50\%$, $a_2 = 4.84\%$, $a_3 = 5.05\%$, $a_4 = 5.13\%$.

Intuition. G&T's rule: when the curve rises, the spot curve is the steepest, the par curve next and the annuity curve the flattest (here spots > par > annuity at years 2–4). The annuity puts weight on early, low-rate years, so its yield is lowest; a par bond puts most weight on the final payment, so its yield is close to, but below, the spot rate. The flat par yield from 3 to 4 years needs a slightly falling spot rate, because the extra coupon is discounted at the higher 3-year rate.

Common mistakes.

- Setting spot yields equal to par yields beyond year 1.
- Discounting the stripped coupons at the par yield instead of at the spot factors already found.
- Averaging spot yields to get the annuity yield; it must be solved as an IRR.

Ref: Grinblatt & Titman Appendix 10A, §10A.1–10A.2 and Examples 10A.1–10A.2, pp. 363–367.

14. The implied value of a call option — Ross 7.32 (DESAFIO), p. 234

What is asked. The value of the issuer's call option embedded in the 8.25% Treasury (callable from February 2008), using two noncallable Treasuries with the same maturity and coupons of 6.5% and 12%.

Approach. The law of one price again. A portfolio of the two noncallable bonds with the right weights has an 8.25% coupon and the same maturity as the callable bond: it is the callable bond *without* the call. The call's value is the price gap (Exercise 4: $P^{\text{callable}} = P^{\text{straight}} - V^{\text{call}}$). Prices are in 32nds of 1% of face (Ross's quote convention).

Notation.

- w: weight on the 6.5% bond, $1 - w$ on the 12% bond (fractions of \$100 face); ask prices 106:12, 103:16, 134:31; bid prices 106:10, 103:14, 134:25.

Step 1 — Match the coupon. Same maturity, so matching coupons matches every cash flow:

$$6.5\,w + 12\,(1-w) = 8.25 \implies 12 - 5.5\,w = 8.25 \implies w = \frac{3.75}{5.5} = 0.681818, \quad 1 - w = 0.318182. \tag{39}$$

Check: $0.681818 \times 6.5 + 0.318182 \times 12 = 4.431818 + 3.818182 = 8.25$. Face values match too: $w + (1 - w) = 1$.

Step 2 — Convert the quotes. $106{:}12 = 106 + 12/32 = 106.375$; $103{:}16 = 103.5$; $134{:}31 = 134 + 31/32 = 134.96875$ (% of face).

Step 3 — Price of the synthetic noncallable 8.25% bond (ask).

$$P^{\text{straight}} = (0.681818)(106.375) + (0.318182)(134.96875) = 72.5284 + 42.9446 = 115.4730. \tag{40}$$

Step 4 — Call value.

$$V^{\text{call}} = P^{\text{straight}} - P^{\text{callable}} = 115.4730 - 103.5 = 11.9730\% \text{ of face} \;\Rightarrow\; \$119.73 \text{ per } \$1{,}000. \tag{41}$$

With bid prices (106.3125, 103.4375, 134.78125): synthetic 115.3707, call = 11.9332%, i.e. \$119.33.

Answer. About \$119.73 per \$1,000 of face value (11.97% of par) using ask prices; \$119.33 using bid prices.

Intuition. The callable bond trades at 103.5 although an otherwise identical noncallable bond is worth 115.5: the Treasury can retire it at par in less than a year, so holders will not pay much above 100 plus the coupons until then. The option is deep in the money (market yields of about 5.25% versus an 8.25% coupon), which is why it is worth about 12% of face.

Common mistakes.

- Reading 106:12 as 106.12; the digits after the colon are 32nds.
- Mixing bid and ask prices across the bonds.
- Comparing yields instead of prices: the callable's quoted yield (5.24%) is not a yield on comparable cash flows.

Ref: Ross §7.2 "Opção de resgate antecipado", p. 207; §7.5 "Relatórios de preços de títulos de dívida", pp. 218–219.

15. A defaultable bond on a binomial tree — Grinblatt & Titman 7.3, p. 254

What is asked. The no-arbitrage value of a two-period (six-month periods) corporate bond paying \$5 at date 1 (secured by a reserve fund) and promising \$105 at date 2, of which only \$50 is paid in the default state DD; risk-free rate 4% per period; asset values per bond as in the exhibit. As an extension tied to block 2A.10 we also compute the bond's promised yield and compare it with its expected return.

Approach. The bond is a derivative on the firm's assets, so G&T's risk-neutral valuation applies: find, at each node, the probability π that makes the assets earn the risk-free rate (eq. 7.4), weight the bond's payoffs with these probabilities, and discount at 4% backwards through the tree. The date-1 coupon is riskless (reserve fund), so its value is $5/1.04$.

Notation.

- $r_f = 4\%$ per period; asset values $S_0 = 100$, $S_U = 120$, $S_D = 90$, $S_{UU} = 300$, $S_{UD} = 110$, $S_{DD} = 50$.
- π_0, π_U, π_D: risk-neutral up-probabilities at nodes 0, U, D; V_U, V_D: date-1 values of the date-2 payment; V_0: bond value.

Step 1 — Risk-neutral probabilities. From $\pi u + (1-\pi)d = 1 + r_f$, i.e. $\pi S_{\text{up}} + (1-\pi)S_{\text{down}} = S(1+r_f)$:

$$\pi_0 = \frac{100(1.04) - 90}{120 - 90} = \frac{104 - 90}{30} = \frac{14}{30} = 0.466667, \tag{42}$$

$$\pi_U = \frac{120(1.04) - 110}{300 - 110} = \frac{124.8 - 110}{190} = \frac{14.8}{190} = 0.077895, \tag{43}$$

$$\pi_D = \frac{90(1.04) - 50}{110 - 50} = \frac{93.6 - 50}{60} = \frac{43.6}{60} = 0.726667. \tag{44}$$

Step 2 — Date-1 values of the date-2 payment. At U both successors pay 105, so the payment is riskless there; at D it is 105 (state UD) or 50 (state DD):

$$V_U = \frac{\pi_U(105) + (1-\pi_U)(105)}{1.04} = \frac{105}{1.04} = 100.9615, \tag{45}$$

$$V_D = \frac{(0.726667)(105) + (0.273333)(50)}{1.04} = \frac{76.30 + 13.6667}{1.04} = \frac{89.9667}{1.04} = 86.5064. \tag{46}$$

Step 3 — Date-0 value. The date-2 payment is worth, at date 0,

$$\begin{aligned} \frac{\pi_0 V_U + (1-\pi_0)V_D}{1.04} &= \frac{(0.466667)(100.9615) + (0.533333)(86.5064)}{1.04} \\ &= \frac{47.1154 + 46.1368}{1.04} = \frac{93.2521}{1.04} = 89.6655, \end{aligned} \tag{47}$$

and the coupon is worth $5/1.04 = 4.8077$. Hence

$$V_0 = 4.8077 + 89.6655 = 94.47. \tag{48}$$

(Same result carrying the coupon through the tree: $[\pi_0(V_U + 5) + (1-\pi_0)(V_D + 5)]/1.04 = 98.2521/1.04 = 94.47$.)

Answer. The bond is worth \$94.47 per \$100 of face value.

Step 4 — Cost of default. A default-free twin would be worth $5/1.04 + 105/1.04^2 = 4.8077 + 105/1.0816 = 101.89$. The risk-neutral probability of reaching DD is $(1-\pi_0)(1-\pi_D) = 0.533333 \times 0.273333 = 0.145778$, and the shortfall there is $105 - 50 = 55$:

$$\frac{55 \times 0.145778}{(1.04)^2} = \frac{8.0178}{1.0816} = 7.4129 = 101.89 - 94.47. \tag{49}$$

Step 5 — Promised yield vs. expected return (block 2A.10). The promised yield y solves $94.47 = 5/(1+y) + 105/(1+y)^2$: $y = 8.10\%$ per half-year (16.21% APR, 16.86% effective). Under the risk-neutral probabilities, the expected return of the bond is exactly $r_f = 4\%$ per period (e.g. at U: $105/100.9615 - 1 = 4\%$). The promised yield exceeds this expected return by $8.10 - 4 = 4.10$ percentage points, all of it compensation for the expected default loss of step 4. Under the true probabilities (not given) the expected return would be 4% plus a risk premium, still below 8.10% because the payoffs can only fall short of the promise.

Intuition. The bond equals a default-free bond minus the value of the shareholders' option to hand over the assets in DD. Its yield measures the promise; its expected return (the cost of debt r_D) is far lower. Using 8.10% as r_D would double the cost of this debt.

Common mistakes.

- Using the same up-probability at every node; the asset returns differ by node ($+20\%/-10\%$ at 0, $+150\%/-8.3\%$ at U), so π differs.
- Discounting the expected payoff under some guessed "true" probability at the risk-free rate; only risk-neutral probabilities go with r_f.
- Taking the promised yield as the bond's expected return.

Ref: Grinblatt & Titman Ch. 7, "Risk-Neutral Valuation of Derivatives", pp. 239–242 (eq. 7.4); Ricca 2A.6–2A.10 (state prices, r_D vs. the yield).